

Blood Pressure Management in the Dialysis Patient

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Hypertension in dialysis patients is common, remains often uncontrolled, and, when diagnosed objectively with out-of-dialysis blood pressure (BP) monitoring, is directly associated with excess risk for mortality. The pathogenesis of hypertension is multifactorial, but volume overload is the most important cause of hypertension in these patients. Accordingly, the initial management relies on nonpharmacologic strategies, such as dry weight adjustment, sodium restriction, and adequate delivery of dialysis time. When BP remains uncontrolled despite aggressive volume management, non-volume-dependent mechanisms are involved as mediators of sustained hypertension (Box 42.1). In such patients, antihypertensive therapy is the next step to achieve BP control.^{1,2} This chapter describes the diagnosis, epidemiology, and management of hypertension among patients on dialysis.

DIAGNOSIS

Peridialytic Blood Pressure Recordings

BP recordings taken shortly before (predialysis) or after dialysis (postdialysis) are the basis for the diagnosis of hypertension among dialysis patients, mainly because these measurements are readily available.^{3,4} However, routine pre- or postdialysis BP recordings provide imprecise estimates of interdialytic ambulatory BP.⁵ Peridialytic BP remains insufficiently accurate, even when a greater number of recordings are averaged over six dialysis sessions.⁶ Thus, BP variability is not the sole factor that accounts for poor diagnostic performance of peridialysis BP recordings.⁷ A potential source of bias could be the lack of attention to proper BP measurement technique. However, even when these measurements are obtained under standardized conditions, peridialytic BP is still inadequate for diagnosing hypertension^{8,9} (Box 42.2).

Intradialytic (During Dialysis) BP Recordings

The diagnostic accuracy of peridialytic BP is improved when these recordings are considered jointly with BP measurements during the entire dialysis procedure¹⁰; the average of intradialytic and peridialytic BP over six consecutive dialysis sessions was superior to predialysis or postdialysis BP alone in predicting interdialytic ambulatory BP.¹⁰ Because calculating an average is impractical at the bedside, this study explored the diagnostic performance of median intradialytic BP from a single midweek dialysis session. A midweek median cut-off systolic BP (SBP) of 140 mm Hg provided 80% sensitivity and 80% specificity in diagnosing hypertension.¹⁰ Therefore, median intradialytic BP can be used to quickly screen for hypertension at the bedside, but this is a method of last resort. Interdialytic BP recordings, though less convenient, may be better for the long-term management of hypertension.

Interdialytic (Between Dialysis) Recordings

Home BP monitoring (HBPM) is an established technique⁴ recommended by guidelines^{11–13} and widely adopted in clinical practice for the

assessment and management of hypertension. This technique improves the diagnosis of hypertension among patients on dialysis.¹⁴ A 1-week average home SBP of 150 mm Hg or above provided 80% sensitivity and 84.1% specificity in diagnosing hypertension. In contrast, no cut-point for routine or standardized peridialytic BP provided a satisfactory combination of sensitivity and specificity for diagnosing hypertension.⁸ Additional advantages of HBPM also exist. In the Dry Weight Reduction in Hypertensive Hemodialysis Patients (DRIP) trial, home BP could track changes in interdialytic ambulatory BP (see later) provoked by probing of dry weight.¹⁵ Contrary to the poor reproducibility of pre- and postdialysis BP, home BP had even greater test-retest reliability than interdialytic ambulatory BP.¹⁵ HBPM also serves as a useful tool to guide therapeutic decisions and facilitate the achievement of adequate BP control. In a randomized trial enrolling 65 hypertensive dialysis patients,¹⁶ compared with predialysis BP-guided management of hypertension, a home BP-guided strategy provoked a reduction of approximately 9/7 mm Hg in interdialytic ambulatory BP.¹⁶ Furthermore, home BP is superior to peridialytic BP in detecting echocardiographic left ventricular hypertrophy (LVH) and in predicting the risk of mortality.^{17–19} For the long-term management of dialysis patients, we recommend twice daily home BP recordings after a midweek dialysis session for 4 days. We tell our patients to keep their home BP monitors in their bedroom and measure BP first thing after they wake up and last thing before going to bed. Monthly monitoring is sufficient for most clinical purposes.

Interdialytic ambulatory BP monitoring (ABPM) is the gold standard technique for diagnosing hypertension.^{3,4} ABPM uses an automated device programmed to record BP 2 to 3 times an hour over the entire interdialytic interval. This method is valid,²⁰ reproducible,²¹ and is superior to peridialytic BP recordings in predicting the presence of LVH or long-term outcomes.^{17–19} A unique advantage of ABPM is that BP can be recorded during sleep, enabling the diagnosis of nocturnal hypertension and nondipping BP patterns.²² ABPM also has some weaknesses (i.e., limited availability and costs of equipment, need for training, patient discomfort) that limit the wide adoption of this technique in practice.^{3,4,22}

Because the implementation of ABPM in the long-term management of hypertension is often impractical, we recommend the use of HBPM as a simple and reliable approach to diagnose hypertension and guide therapeutic decisions from month to month.^{1,2} In patients who are nonadherent to HBPM, we use midweek median intradialytic BP to make therapeutic decisions.^{1,2} We do not use peridialytic BP recordings for managing hypertension.

EPIDEMIOLOGY

Prevalence, Treatment, and Control of Hypertension

Epidemiology of hypertension among patients on dialysis differs depending on the method of BP measurement (Table 42.1). Studies

BOX 42.1 Non-Volume-Dependent Causes of Hypertension Among Patients on Dialysis

- Increased arterial stiffness
- Sympathetic overactivity
- Activation of the renin-angiotensin-aldosterone system
- Endothelial dysfunction
- Erythropoietin therapy
- Sleep apnea

that were based on conventional predialysis BP recordings used different definitions and provided variable estimates of the prevalence and control of hypertension.²³⁻²⁶ A more precise description of the epidemiology was provided by a few studies using ABPM. For example, in a cross-sectional study of 369 patients,²⁷ the prevalence of hypertension (defined as interdialytic BP $\geq 135/85$ mm Hg or antihypertensive drug use) was 82%. Although 89% of hypertensive dialysis patients were being treated, adequate ambulatory BP control was achieved in only 38%.²⁷ Subsequent studies using HBPM or ABPM confirmed that the burden of hypertension in the dialysis population is high and its control poor.^{28,29}

Among patients on peritoneal dialysis, the absence of acute shifts in pressure and volume parameters is thought to mitigate hypertension. However, the burden of hypertension and prognostic association of BP with mortality follows a pattern similar to that seen among hemodialysis.³⁰

Prognosis

In sharp contrast with the direct and linear association between BP and outcome in the general population, large-scale cohort studies conducted in dialysis patients showed a U-shaped or J-shaped association of peridialytic BP with mortality.³¹⁻³⁴ The phenomenon of reverse epidemiology has raised concerns on whether hypertension in dialysis patients is an independent risk factor that should be adequately controlled.³⁵ However, two separate cohort studies showed that BP measurement technique strongly confounds the relation of BP with mortality.^{18,19} In these studies, increasing interdialytic SBP assessed either with HBPM or with ABPM was directly associated with higher mortality risk. In contrast, neither routine nor standardized peridialytic BP recordings were prognostically informative. Home SBP between 120 to 130 mm Hg and 44-hour ambulatory SBP ranging from 110 to 120 mm Hg were associated with the best prognosis.¹⁸

The reverse epidemiology of hypertension seen in observational studies contrasts with two meta-analyses of randomized trials among dialysis patients showing cardiovascular protection with antihypertensive therapy.^{36,37} In the first meta-analysis (eight trials, $N = 1679$), BP-lowering treatment was associated with a 29% reduced risk for cardiovascular events and a 20% reduction in all-cause mortality.³⁷ In the second meta-analysis (five trials, $N = 1212$), compared with placebo, antihypertensive therapy lowered the risk for cardiovascular events and all-cause mortality by 31%.³⁶ The cardiovascular benefit of antihypertensive therapy was greater when trials with hypertensive patients were combined separately from trials that included normotensives.³⁶ Therefore, deliberate BP lowering with antihypertensive therapy has proven cardioprotective benefits for dialysis patients, particularly when they are hypertensive.

THERAPEUTIC TARGETS

The optimal BP threshold for the diagnosis and management of hypertension in the dialysis population is unknown.^{12,38} The only trial to

BOX 42.2 Blood Pressure Measurement Techniques in Dialysis Patients

- Predialysis or postdialysis (peridialysis) BP: do not reflect the usual level of BP in the interdialytic period and therefore not useful for management of hypertension.
- Median intradialytic BP: systolic BP ≥ 140 mm Hg on a midweek dialysis session in a patient being dialyzed 3 times weekly suggests poorly controlled hypertension.
- Home BP monitoring: measurements done twice daily (morning and evening) after 5 min of seated rest in duplicate, after a midweek dialysis session for 4 days on a monthly basis, is the most practical way to manage hypertension in long-term hemodialysis patients.
- Interdialytic ambulatory BP monitoring: measurements after dialysis 2–3 times/hr over the entire 44-hr interdialytic period is the gold standard of diagnosing hypertension in hemodialysis patients.

BP, Blood pressure.

compare two different BP targets was the Blood Pressure in Dialysis (BID) trial.³⁹ In BID, 126 hypertensive dialysis patients were randomized to a tighter (110–140 mm Hg) versus a standard (155–165 mm Hg) control of predialysis SBP over 12 months of follow-up. BID reported a possible safety signal, but on closer examination intensive BP lowering was associated with 1.3-fold higher risk for recurrent intradialytic hypotension, 1.66-fold higher risk for recurrent hospitalization, and 2.8-fold higher risk for recurrent vascular access thrombosis.³⁹ Regression of LVH, a secondary efficacy endpoint, did not differ between arms.

BID was a feasibility trial aiming to inform the design of a full-scale trial.³⁹ Although a full-scale trial is warranted to define the optimal BP goals, we do not believe that peridialytic BP recordings are appropriate targets for managing hypertension. Given the evidence supporting the superiority of home over predialysis BP-guided therapy,^{14,16} we believe that a future full-scale trial should ideally compare the benefits of tighter versus standard home SBP targets for clinical outcomes. In the meantime, we recommend that controlling home SBP to levels below 140 mm Hg is a reasonable target in dialysis patients.^{1,2} If a home BP-guided strategy is not feasible, lowering midweek median intradialytic SBP below 140 mm Hg is an alternative strategy.^{1,2} In our practice, we prefer to treat hypertension in dialysis patients than maintain the status quo.

VOLUME MANAGEMENT STRATEGIES

Once an accurate diagnosis is made, the initial management of hypertension is based on nonpharmacologic strategies that target and maintain euvolemia.^{1,2} The four basic principles of nonpharmacologic treatment are summarized in Fig. 42.1 and are discussed in detail later.

Probing of Dry Weight

The assessment and management of dry weight is challenging. There is no consensus on the optimal definition of dry weight. In 2009 Sinha and Agarwal stated that dry weight reflects the lowest tolerated postdialysis weight at which the patient experiences minimal signs and symptoms of either hypovolemia or hypervolemia.⁴⁰ According to this definition, the management of dry weight is based on an iterative process of gentle and gradual intensification of ultrafiltration guided by the patient's symptoms.⁴⁰

Benefits of Probing Dry Weight

Dry weight was probed without increasing the dialysis duration in the DRIP trial.⁴¹ In DRIP, 150 dialysis patients who had uncontrolled hypertension despite stable background treatment with 2.7 antihypertensive

TABLE 42.1 Prevalence, Treatment and Control of Hypertension Among Patients on Dialysis

Study	Region	N	Definition of Hypertension	Prevalence	Hypertensive Patients on Treatment	Control
Studies Using Dialysis Unit BP Recordings						
Salem, ²⁶ 1995	US	649	Predialysis mean BP $\geq$ 114 mm Hg or antihypertensive drug use	71.9%	81.5%	48.6%
Rahman et al., ²⁵ 1999	US	489	Predialysis BP $\geq$ 160/90 mm Hg or antihypertensive drug use	87.7%	92.3%	28.9%
Agarwal et al., ²³ 2003	US	2535	Predialysis BP $\geq$ 150/85 mm Hg or antihypertensive drug use	85.7%	88.4%	30.3%
Del Vecchio et al., ²⁴ 2013	Italy	4022	Predialysis BP $\geq$ 140/90 mm Hg or antihypertensive drug use	70.3%	57.7%	40.0%
Studies Using Home or Ambulatory BP Monitoring						
Agarwal, ²⁷ 2011	US	369	44-hour interdialytic BP $\geq$ 135/85 mm Hg or antihypertensive drug use	82.0%	89.0%	38.0%
Sarafidis et al., ²⁸ 2019	Italy, Greece, Slovenia	396	48-hour intra- and interdialytic BP $\geq$ 130/80 mm Hg or antihypertensive drug use	84.3%	86.8%	28.7%
Tsikliras et al., ²⁹ 2020	Greece	116	1-week averaged home BP $\geq$ 135/85 mm Hg or antihypertensive drug use	88.8%	95.0%	32.6%

BP, Blood pressure.

Management of Hypertension Among Patients on Dialysis

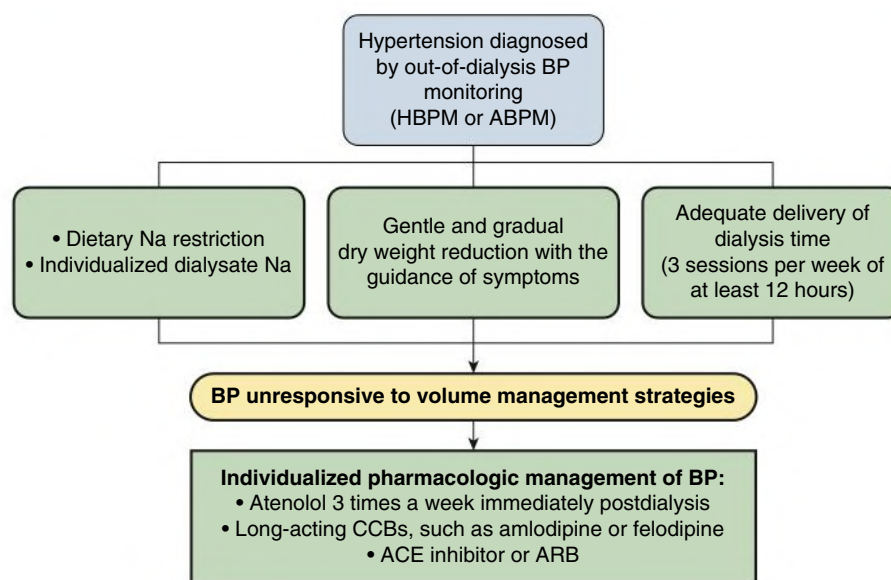


Fig. 42.1 Algorithm for the Management of Hypertension Among Patients on Dialysis. The diagnosis of hypertension among patients on dialysis should be based on blood pressure (BP) recordings taken outside of dialysis. Predialysis and postdialysis BP recordings are not suitable for diagnosing and managing hypertension. Once an accurate diagnosis is made, first-line management of hypertension among patients on dialysis relies on nonpharmacologic strategies that target the achievement and maintenance of euvolemia. The adequate management of volume follows four principles: probing of dry weight, dietary sodium restriction, individualized prescription of the dialysate sodium concentrations, and adequate delivery of dialysis time. If hypertension remains uncontrolled despite adequate volume management, pharmacologic treatment is the next step to achieve an adequate control of BP. The selection of an appropriate antihypertensive regimen should be individualized, taking into consideration the efficacy, safety, and pharmacokinetic properties of the antihypertensive medications in the dialysis population. We use β -blockers, particularly atenolol administered 3 times per week immediately postdialysis, as first-line therapy. We use long-acting calcium channel blockers (CCBs), such as amlodipine or felodipine, as second-line therapy in patients with uncontrolled BP despite the adequate management of dry weight and the administration of atenolol. Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) are our third-line option. Because of concerns related to hyperkalemia and in anticipation of stronger safety and efficacy data from ongoing randomized trials, we do not recommend the wide use of mineralocorticoid receptor antagonists for the management of hypertension among patients on dialysis. *ABPM*, Ambulatory blood pressure monitoring; *HBPM*, home blood pressure monitoring; *Na*, sodium.

Ambulatory BP Reduction Over 4 and 8 Weeks of Dry Weight Probing

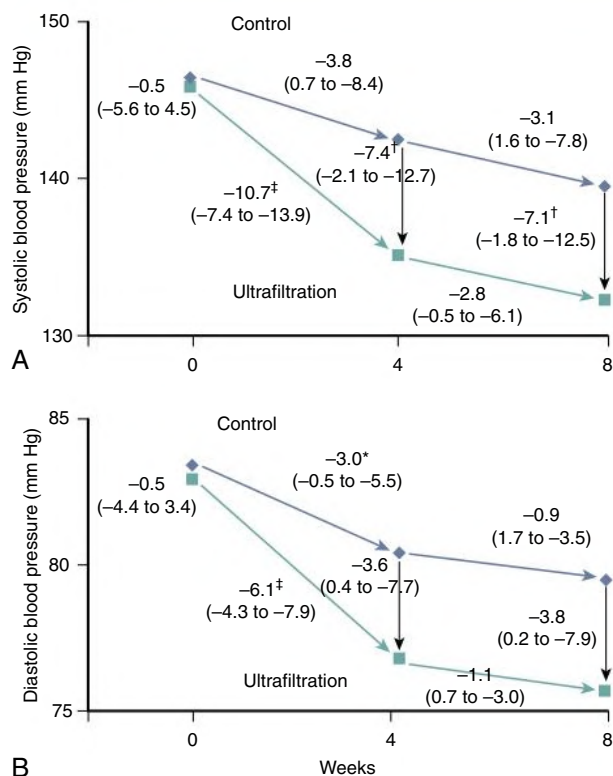


Fig. 42.2 Ambulatory Blood Pressure (BP) Reduction Over 4 and 8 Weeks of Dry Weight Probing in Hypertensive Dialysis Patients. The mean 44-hour ambulatory systolic and diastolic BPs are shown for the control (blue lines) and ultrafiltration (teal lines) groups. The mean changes in BP are shown for weeks 4 and 8 after randomization (teal squares and arrows), as well as the mean differences in BPs (black arrows) between the two groups at each 4-week interval. The numbers next to the blue diamonds and arrows connecting the data points are the mean changes in BP between groups at 4 and 8 weeks after randomization. The 95% confidence intervals (CIs) are given in parentheses. * $P < .05$, † $P < .01$, and ‡ $P < .001$ indicate significant differences between groups or within groups. The placebo-subtracted change in 44-hour systolic BP provoked by probing of dry weight was -6.9 mm Hg (95% CI, -12.4 to -1.3 mm Hg; $P = .016$) at 4 weeks and -6.6 mm Hg (95% CI, -12.2 to -1.0 mm Hg; $P = .021$) at 8 weeks. The placebo-subtracted change in 44-hour diastolic BP was -3.1 mm Hg (95% CI, -6.2 to -0.02 mm Hg; $P = .048$) at 4 weeks and -3.3 mm Hg (95% CI, -6.4 to -0.2 mm Hg; $P = .037$) at 8 weeks.

medications daily were randomized in a 2:1 ratio to ultrafiltration and control groups. In the ultrafiltration group, postdialysis weight was gradually reduced until the development of symptoms indicating dry weight achievement. In the control group, patients received only physician visits.⁴¹ A modest reduction in dry weight by 0.9 kg from baseline to 4 weeks provoked an average placebo-subtracted reduction of -6.9 mm Hg in 44-hour SBP and a placebo-subtracted reduction of -3.1 mm Hg in 44-hour diastolic BP (DBP) (Fig. 42.2).⁴¹ This BP-lowering effect was sustained till study end at 8 weeks. Probing of dry weight also provoked a significant reduction in LV mass index (LVMI) over the first 4 weeks of follow-up.⁴²

Assessment and Management of Dry Weight

The results of the DRIP trial⁴¹ have important implications for the assessment and management of dry weight (Box 42.3).

BOX 42.3 Clinical Signs Indicating Volume Excess in a Dialysis Patient

- Overt symptoms of congestive heart failure (e.g., shortness of breath)
- Uncontrolled hypertension confirmed by home or ambulatory BP monitoring
- The use of multiple antihypertensive medications
- Low interdialytic weight gain
- Persistent BP elevation during dialysis (intradialytic hypertension)

BP, Blood pressure.

The DRIP trial explored the hypothesis that hypertension in dialysis is a clinical manifestation of volume excess. Accordingly, patients with clinically overt hypervolemia were a priori excluded from this trial.⁴⁰ Despite the fact that DRIP participants were euvolemic based on clinical judgment, probing of dry weight provoked a clinically meaningful reduction of approximately 7/3 mm Hg in 44-hour BP.⁴¹ Thus, among hypertensive dialysis patients, the management of dry weight should not be based on the presence or absence of clinically overt hypervolemia. In such patients, volume excess is more often covert. The discriminatory power of physical examination in ruling in or out volume overload is low.^{43,44}

An important sign that should raise the suspicion of volume excess is the use of multiple BP-lowering medications in a dialysis patient. DRIP participants had uncontrolled hypertension despite the administration of 2.7 BP-lowering medications daily.⁴¹ The mediator of sustained hypertension in these patients was subclinical volume excess. In DRIP, the magnitude of ambulatory BP reduction evoked by probing of dry weight was probably much larger than what would be expected by intensifying antihypertensive therapy with one additional drug. This is in line with several cross-sectional studies showing an increasing number of prescribed antihypertensive medications to be paradoxically associated with worse BP control.^{23,27,29} These observations suggest that intensification of antihypertensive therapy is likely to fail to control BP if volume excess is not primarily addressed, supporting a “volume-first” approach of hypertension.

A high interdialytic weight gain (IDWG) is often (but erroneously) considered as equivalent with hypervolemia. However, recent studies using more objective methods in the assessment of dry weight inform us that IDWG and volume overload are two discrete components of the dynamic fluid balance.⁴⁴ These studies showed that dialysis patients with low IDWG were often volume overloaded.⁴⁴ In DRIP,⁴⁵ volume status was assessed using the method of relative plasma volume (RPV) monitoring. At baseline, patients with steeper RPV slopes and therefore most euvolemic had the highest IDWG. In contrast, patients with the greatest volume expansion based on RPV monitoring had the lowest IDWG.⁴⁵ However, these patients tolerated the greatest dry weight reduction during follow-up (-1.5 kg). Most importantly, these patients experienced the greatest improvement in 44-hour SBP in response to dry weight reduction (-12.6 mm Hg).⁴⁵ Therefore, a low IDWG may reflect volume excess, particularly when accompanied by other clinical indications, such as uncontrolled interdialytic hypertension.

Unlike the typical decline in BP with ultrafiltration, BP increases during dialysis in approximately 10% to 15% of patients. This paradoxical hemodynamic response is described as *intradialytic hypertension*.⁴⁶ The precise definitions and pathophysiology of intradialytic hypertension is debated; earlier studies suggested that this phenomenon may be mediated through release of vasoconstrictors in response to ultrafiltration.⁴⁶ On this basis, the management of intradialytic hypertension often relied on interruption of ultrafiltration and/or immediate administration of potent short-acting antihypertensive agents. However, more recent studies showed a close relation between

TABLE 42.2 Randomized Trials Comparing Management of Dry Weight Guided by Assistive Technologies Versus Usual Care on Clinical Outcomes

Study	Design	Patients	N	Intervention	Follow-up	Results
Reddan et al. ⁵⁸	Open-label RCT	Patients on dialysis (≥2 mo)	443	RPV monitoring using Crit-Line vs. conventional clinical monitoring	6 mo	Higher risk of hospitalization and all-cause mortality in the RPV monitoring group than in the control group
Onofriescu et al. ⁵⁷	Open-label RCT	Patients on dialysis (≥3 mo)	131	BIS every 3 mo to guide dry weight adjustment vs. clinical judgment	30 mo	One death in the BIS group vs. 8 deaths in the control group (HR, 0.112; 95% CI, 0.014–0.918)
Huan-Sheng et al. ⁵⁵	Open-label RCT	Patients on dialysis (≥3 mo)	298	BIS every month to guide dry weight adjustment vs. clinical judgment	12 mo	No between-group difference in all-cause hospitalizations (HR, 1.19; 95% CI, 0.79–1.80), cardiovascular events (HR, 0.57; 95% CI, 0.26–1.25), and all-cause mortality (HR, 0.85; 95% CI, 0.29–2.53)
Siriopol et al. ⁵⁹	Open-label RCT	Patients on dialysis (≥3 mo) at low cardiovascular risk	250	Lung ultrasound combined with BIS (in case of clinical symptoms of hypovolemia) vs. clinical judgment	Mean 21.3 ± 5.6 mo	No between-group difference in the composite outcome of all-cause death or first nonfatal cardiovascular event (HR, 1.09; 95% CI, 0.64–1.86)
Liu et al. ⁵⁶	Open-label RCT	Patients on dialysis (≥3 mo)	445	BIS every 2 months to guide dry weight adjustment vs. clinical judgment	Median 13.7 mo	No between-group difference in the composite outcome of all-cause death or nonfatal cardiovascular event (HR, 0.487; 95% CI, 0.217–1.091)

BIS, Bioimpedance spectroscopy; CI, confidence interval; HR, hazard ratio; RCT, randomized controlled trial; RPV, relative plasma volume.

intradialytic and interdialytic hypertension.⁴⁷ In a post hoc analysis of the DRIP trial, probing of dry weight in patients with intradialytic hypertension was effective in normalizing both intradialytic and interdialytic BP profiles.⁴⁸ Therefore, persistent BP elevation during dialysis may be another signal of volume excess.

Bioimpedance spectroscopy, RPV monitoring, and lung ultrasound, are under investigation and offer promise for more objective assessment of dry weight.^{45,49,50} Observational studies showed that hypervolemia, as assessed with these devices, is prognostically associated with higher mortality risk.^{51–54} Randomized trials have so far failed to prove that these technologies are superior to usual care in improving clinical outcomes (Table 42.2),^{55–59} although most of these trials had short follow-up and lacked statistical power.^{60,61} Therefore, these technologies require further study before being implemented in clinical practice.

Potential Hazards of Probing Dry Weight

Probing of dry weight may be associated with potential hazards, such as higher risk for intradialytic hypotension, vascular access thrombosis, and more rapid loss of residual renal function.^{1,2} However, none of these risks has been adequately studied. In DRIP, probing of dry weight provoked temporal intradialytic symptoms of hypotension, but these symptoms did not unduly affect any domain of health-related quality of life.⁴¹ Because DRIP only followed participants for 8 weeks, the benefits and risks of dry weight probing warrant examination in long-term trials.

Dietary Sodium Restriction

Dietary sodium restriction is an established nonpharmacologic approach to limit IDWG and facilitate the achievement of dry weight.^{1,2} International guidelines recommend that among patients on dialysis, dietary sodium intake should not exceed 80 to 100 mmol (1.8–2.3 g; equivalent to 4.5–5.8 g sodium chloride) daily.⁶² Observational studies support this guidance, showing that compared with pharmacologic management of hypertension, sodium restriction together with adequate adjustment in dry weight was associated with greater regression of LVH and reduced incidence of intradialytic hypotension.^{63,64}

Individualized Dialysate Sodium Prescription

Sodium loading in hemodialysis patients occurs when the prescription of dialysate sodium concentration results in a positive sodium gradient during dialysis.⁶⁵ This scenario is rather common in daily clinical practice.⁶⁶ However, this therapeutic approach perpetuates a vicious cycle. Intradialytic sodium gain is directly associated with increased sense of thirst and greater IDWG, resulting in higher ultrafiltration requirements during the subsequent dialysis sessions. The higher ultrafiltration rates can aggravate the risk of intradialytic hypotension, which might result in the prescription of even higher dialysate sodium concentrations.^{1,2}

Lowering the dialysate sodium concentration could possibly interrupt this vicious cycle. The benefit/risk ratio of this intervention was investigated in a meta-analysis of 12 trials involving 266 patients.⁶⁷ Compared with neutral (138–140 mEq/L) or high (>140 mEq/L)

dialysate sodium, a low sodium concentration (<138 mEq/L) was associated with reduced IDWG and improvement in BP. However, these benefits were accompanied by higher risk for intradialytic cramps and hypotension.⁶⁷ These results should be carefully interpreted because all trials were small. Of the 12 trials included in this meta-analysis, 10 compared a fixed prescription of a lower versus a higher dialysate sodium concentration. Several trials performed large reductions in dialysate sodium concentration over a short-term follow-up.⁶⁷ Therefore, an abrupt and fixed intervention may result in a steeply negative sodium gradient, raising the risk for adverse events.

We recommend that lowering of dialysate sodium concentration should be gradual and individualized.^{1,2} Observational studies and pilot trials suggest that this approach has the potential for short-term and long-term benefits without excess risk of intradialytic complications.⁶⁸⁻⁷⁰

Adequate Delivery of Dialysis Time

The prescription of dialysis duration varies considerably across countries. Approximately one-third of patients in the United States are prescribed hemodialysis with a duration of less than 200 minutes.⁷¹ Among several other risks, nonadherence to the dialysis regimen has been associated with worse BP control.⁷² This was also illustrated in a post hoc analysis of DRIP.⁷³ Median intradialytic SBP followed a rising pattern in the control group. In contrast, probing of dry weight lowered median intradialytic SBP regardless of the duration of dialysis. However, patients with shorter dialysis duration required more dialysis sessions to achieve the BP-lowering benefit of probing dry weight.⁷³ Therefore, shorter delivered dialysis is a barrier to dry weight achievement, limiting the opportunity for adequate BP control.

Management of Hypertension in Peritoneal Dialysis Patients

As in hemodialysis patients, volume management is of central importance for BP control among peritoneal dialysis (PD) patients.³⁰ Two unique aspects in PD patients are long-term preservation of both residual diuresis and peritoneal membrane function. To achieve euvoolemia, we adapt the PD regimen to the peritoneal transport characteristics. To protect the peritoneal membrane, we limit exposure of peritoneal membrane to bioincompatible solutions and use icodextrin judiciously³⁰ (prescription of PD is discussed in Chapter 101).

PHARMACOTHERAPY OF HYPERTENSION (BASED ON INTERDIALYTIC MEASUREMENTS)

Despite assiduously adhering to the volume-first management strategies, a large fraction of dialysis patients remain hypertensive, and drug therapy is necessary to achieve an adequate control of BP.

We believe that treatment should be guided by evidence from trials done in the dialysis population and not on extrapolation of evidence from trials conducted in earlier stages of kidney disease or in the general population.^{1,2} A summary of key randomized trials is provided in Table 42.3.⁷⁴⁻⁸²

β-Blockers

The safety and efficacy of β-blocker-based and angiotensin-converting enzyme (ACE) inhibitor-based regimens were compared in the Hypertension in Hemodialysis Patients Treated With Atenolol or Lisinopril (HDPAL) trial.⁷³ In HDPAL, 200 hypertensive dialysis patients with echocardiographic LVH were randomized to atenolol or lisinopril, each administered 3 times per week immediately postdialysis. Contrary to the original hypothesis that a lisinopril-based regimen would be superior to atenolol in improving LVH, both drugs

BOX 42.4 Dialyzability and Dosing of Antihypertensive Agents^{2,73,82,83}

- β-blockers are a heterogeneous drug class with respect to their pharmacokinetic profiles. Recent studies showed substantial dialyzability of atenolol, metoprolol, and bisoprolol; in contrast, carvedilol and propranolol were found to be nondialyzable.
- With the exception of the nondialyzable fosinopril, most other ACE inhibitors are removed during dialysis; in contrast, ARBs are generally nondialyzable.
- Long-acting CCBs, such as amlodipine and felodipine, are also nondialyzable and can be dosed once daily.
- Based on the HDPAL trial, the optimal dosing regimen for long-acting agents with high dialyzability, such as atenolol and lisinopril, is the administration 3 times per week immediately postdialysis.

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CCB, calcium channel blocker; HDPAL, Hypertension in Hemodialysis Patients Treated With Atenolol or Lisinopril.

provoked similar reductions in LVMI over 12 months.⁷⁴ In addition, atenolol appeared to be more effective in controlling hypertension, because monthly monitored home BP was consistently lower in the atenolol group despite the need for a greater number of antihypertensive drugs and a greater dry weight reduction in the lisinopril group. Furthermore, compared with lisinopril, atenolol provoked a greater reduction in aortic pulse wave velocity over the first 6 months of follow-up.⁸³ Because the incidence of serious adverse cardiovascular events was 2.36-fold higher in the lisinopril group than in the atenolol group, HPDAL was terminated early.⁷⁴

Based on the results of the HDPAL trial, we use β-blockers as first-line pharmacotherapy for hypertension among dialysis patients (Fig. 42.1)^{1,2} and prefer to use atenolol administered 3 times per week immediately postdialysis based on the results of HDPAL. This dosing regimen overcomes the high dialyzability of atenolol and ensures a potent BP-lowering action of this agent over the entire interdialytic interval (Box 42.4).⁸⁴

Calcium Channel Blockers

Long-acting dihydropyridine calcium channel blockers (CCBs) such as amlodipine or felodipine are useful in combination with other agents, such as atenolol, and represent our second-line choice for the pharmacotherapy of hypertension in dialysis patients. The efficacy of this drug class is supported by a double-blind trial in which 251 hypertensive dialysis patients were randomized to amlodipine or placebo.⁸⁰ Whereas BP remained constant with placebo, amlodipine provoked a significant reduction of ~10 mm Hg in SBP over 30 months of follow-up. Although amlodipine did not reduce all-cause mortality relative to placebo, there was a significant reduction by 47% in the composite secondary outcome of cardiovascular events and all-cause mortality.⁸⁰

ACE Inhibitors/Angiotensin Receptor Blockers

In the Fosinopril in Dialysis (FOSIDIAL) trial,⁸¹ 397 French dialysis patients who had echocardiographic LVH with or without hypertension were randomized to the ACE inhibitor fosinopril or placebo for 48 months. Compared with placebo, fosinopril had no benefit on cardiovascular outcomes.⁸¹ In contrast, two smaller trials from Japan showed cardioprotective benefit when an angiotensin receptor blocker (ARB)-based regimen was compared either with placebo⁷⁹ or with active treatment that was not based on ACE inhibitors or ARBs.⁷⁸ However, in a subsequent meta-analysis aiming to elucidate the contradictory results of these three small trials, the use of ACE inhibitors or ARBs was not

TABLE 42.3 Randomized Trials Evaluating the Effect of Antihypertensive Agents on Clinical Outcomes

Study	Design	Patients	N	Intervention	Follow-up	Results
Zannad et al. ⁸¹	Double-blind RCT	Dialysis patients (not necessarily hypertensive) with LVH	397	Fosinopril 5–20 mg/day vs. placebo	24 mo	No benefit of fosinopril vs. placebo on the combined outcome of fatal or nonfatal CV events (RR, 0.93; 95% CI, 0.68–1.26)
Takahashi et al. ⁷⁹	Open-label RCT	Dialysis patients with clinical evidence of CV disease	80	Candesartan 4–8 mg/day vs. nothing	19 mo	Threefold lower incidence of CV events in the candesartan group than in the control group (45.9% vs. 16.3%, $P < .01$)
Suzuki et al. ⁷⁸	Open-label RCT	Hypertensive dialysis patients	366	ARB treatment (losartan, valsartan, candesartan) vs. control therapy not based on ARBs	36 mo	Compared with control therapy, treatment with ARBs improved by 49% the combined outcome of fatal and nonfatal CV events (HR, 0.51; 95% CI, 0.33–0.9)
Iseki et al. ⁷⁵	Open-label RCT	Hypertensive dialysis patients	469	Olmesartan 10–40 mg/day vs. control therapy not including ACE inhibitor or ARB	42 mo	No benefit of olmesartan vs. control therapy on the combined outcome of fatal or nonfatal CV events (HR, 1.00; 95% CI, 0.71–1.40)
Cice et al. ⁸²	Open-label RCT	Dialysis patients with dilated cardiomyopathy	114	Carvedilol (titrated up to 25 mg twice daily) vs. placebo	24 mo	Compared with placebo, carvedilol improved all-cause mortality (HR, 0.51; 95% CI, 0.32–0.82) and CV mortality (HR, 0.32; 95% CI, 0.18–0.57)
Agarwal et al. ⁷⁴	Open-label RCT	Hypertensive dialysis patients with LVH	200	Atenolol (25–100 mg) vs. lisinopril (10–40 mg), both administered 3 times a week postdialysis	12 mo	Compared with atenolol, the composite safety outcome of MI, stroke, or CHF occurred more commonly in the lisinopril group (IRR, 2.36; 95% CI, 1.36–4.23)
Tepel et al. ⁸⁰	Double-blind RCT	Hypertensive dialysis patients	251	Amlodipine 10 mg/day vs. placebo	30 mo	Amlodipine improved the composite outcome of all-cause death or nonfatal CV event relative to placebo (HR, 0.53; 95% CI, 0.31–0.93)
Matsumoto et al. ⁷⁷	Open-label RCT	Oligoanuric dialysis patients	309	Add-on spironolactone 25 mg/day vs. nothing	36 mo	Compared with no add-on therapy, spironolactone reduced the composite outcome of fatal and nonfatal CV events (HR, 0.379; 95% CI, 0.173–0.832)
Lin et al. ⁷⁶	Open-label RCT	Patients on dialysis without CHF	253	Add-on spironolactone 25 mg/day vs. placebo	24 mo	Add-on spironolactone improved CV morbidity and mortality relative to placebo (HR, 0.42; 95% CI, 0.26–0.78)

ACE, Angiotensin-converting enzyme; ARB, angiotensin receptor blocker; CHF, congestive heart failure; CI, confidence interval; CV, cardiovascular; HR, hazard ratio; IRR, incident rate ratio; LVH, left ventricular hypertrophy; MI, myocardial infarction; RCT, randomized controlled trial; RR, risk ratio.

associated with an overall improvement in cardiovascular morbidity and mortality.⁸⁵ In the Olmesartan Clinical Trial in Okinawa Patients under Okinawa Dialysis Study (OCTOPUS) trial⁷⁵—the largest trial conducted so far, enrolling 469 hypertensive dialysis patients—the ARB olmesartan was not superior to control for improving the primary composite cardiovascular outcome or all-cause mortality over a mean follow-up of 3.5 years.⁷⁵

Taken together, unlike the established cardioprotective benefit of these two antihypertensive drug classes in the general population, the evidence to support the use of ACE inhibitors or ARBs for cardioprotection in patients on dialysis is thin. Therefore, we use ACE inhibitors and ARBs as a third-line option in dialysis patients with inadequately controlled BP despite combination treatment with atenolol and a long-acting CCB.

Mineralocorticoid Receptor Agonists

Among patients not on dialysis with heart failure and reduced ejection fraction, add-on therapy with spironolactone or eplerenone confers a substantial cardioprotective benefit.^{86,87} However, the safety and efficacy of mineralocorticoid receptor antagonists (MRAs) among patients on dialysis has not been adequately studied.^{76–88} In a 2016 meta-analysis of nine trials (including 829 patients), MRA use was associated with improvement in cardiovascular and all-cause mortality, but this benefit was accompanied by a threefold higher risk of hyperkalemia.⁸⁹ These results should be interpreted with caution because most of the included trials had short follow-up and small sample sizes, indicating lack of high-quality evidence.⁸⁹ The benefit/risk ratio of MRAs in patients on dialysis is currently under investigation

in two large ongoing phase III trials. Because of concerns related to hyperkalemia and in anticipation of more reliable evidence, we do not routinely prescribe MRAs to our dialysis patients.

Vasodilators

Many patients on dialysis are prescribed vasodilators such as hydralazine or minoxidil. Hydralazine is short acting, which is why it needs to be administered 3 times daily. In reviewing ABPM data on patients on this drug, we often note large drops in SBP with rapid increases to baseline; we therefore discourage its use. Minoxidil prescription is associated with hirsutism that is troublesome for women and sometimes can cause pericardial effusions. Patients who require a vasodilator to achieve blood pressure control typically will benefit from nonpharmacologic strategies for the control of hypertension.

Role of Bilateral Nephrectomy

We have rarely seen patients with bilateral nephrectomies as a strategy to treat hypertension in dialysis. In one such patient, BP was still elevated several months after surgery and the patient was being administered an intravenous agent to control hypertension in the intensive care unit. Two days of continuous venovenous hemofiltration normalized the BP and required removal of several antihypertensive drugs including the intravenous agent, attesting to the fact that the optimal control of volume and an adequate delivery of dialysis are central to the management of hypertension in most hemodialysis patients. Instead of an as-needed prescription of an antihypertensive drug for the patient with asymptomatic elevation of BP to 200 mm Hg systolic, we prefer to address volume and dialysis delivery for the holistic care of the patient.

SELF-ASSESSMENT QUESTIONS

- Which of the following methods is useful to screen for hypertension in a dialysis patient at bedside?
 - Predialysis BP recordings averaged over six consecutive dialysis treatments.
 - Postdialysis BP recordings averaged over six consecutive dialysis treatments.
 - Median intradialytic BP from a single midweek dialysis treatment.
 - None of the above methods provides a satisfactory combination of high sensitivity and high specificity for diagnosing hypertension.
- A 60-year-old patient on chronic dialysis with a known history of hypertension receives stable treatment with nebivolol 5 mg/day, amlodipine 10 mg/day, and olmesartan 20 mg/day, but the average of home BP recordings taken with a validated automatic device twice daily for 4 days after the midweek dialysis session is 149/78 mm Hg. The patient is asymptomatic, and the physical examination reveals no clinical signs of volume excess. Which of the following would be the optimal approach to achieve an adequate control of home BP in this patient?
 - Increase the dose of olmesartan
 - Add spironolactone
 - Add clonidine
 - Probe dry weight
- In a dialysis patient who gains 1 kg during the interdialytic interval, the predialysis BP is of 154/82 mm Hg. During dialysis, BP increases to 180/89 mm Hg, and the nurse calls you with this high BP. Which of the following interventions would be the *best* next step to treat the patient?
 - Stop ultrafiltration immediately and administer 10 mg sublingual nifedipine
 - Stop ultrafiltration
 - Administer clonidine 0.3 mg orally
 - Probe dry weight

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